



Review Article

Zebrafish models for studying central nervous system effects of Ayurvedic medicinal plants

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ABSTRACT

Traditional Indian medicine (*Ayurveda*) has a long history of treating central nervous system (CNS) disorders. *Ayurvedic* plants are typically rich in flavonoids, lignans, sterols, tannins and alkaloids that exert potent anti-inflammatory, anti-oxidant, neuroprotective and neurotropic effects in both humans and rodents. The zebrafish (*Danio rerio*) has emerged as a useful model species and a powerful vertebrate in vivo system for CNS disease modeling and drug screening. As zebrafish continue to demonstrate their growing utility for studying a wide range of pharmaceutical therapies, including Chinese and American traditional medicines, here we discuss CNS effects of various *Ayurvedic* medicinal plants, and how they can be further elucidated using zebrafish models. We also evaluate the existing challenges and limitations of using zebrafish models for studying *Ayurvedic* pharmacotherapy, as well as outline future directions of translational research in this field.

1. Introduction

Traditional Indian medicine, *Ayurveda*, has been in use for more than 5000 years [1], commonly applied in the region to treat various human diseases, including cancer, infectious, metabolic, cardiovascular and neurological disorders [2]. *Ayurveda* (meaning “The Science of Life” in *Sanskrit*) uses native plants rich in flavonoids, lignans, sterols, tannins and alkaloids, with robust anti-inflammatory, anti-arthritis, antioxidant and other beneficial properties [3,4]. Today, *Ayurveda* remains one of the oldest and most commonly used therapies globally, whose popularity is also expanding. For example, the *Ayurvedic* medicines have a long history of use and relatively few side effects, hence contributing to its growing consumption, with nearly 90 % Indians and up to 10 %

American adults presently treated by *Ayurvedic* medicines.

Numerous *Ayurvedic* plants (e.g., curcumin, Indian pennywort, Indian ginseng, tulsi) have long been used to treat various brain disorders [4], known as ‘*vata-vyadhi*’ in *Sanskrit* (from ‘*vata*’ - energy moving from brain to nerves to control human body, and ‘*vyadhi*’, the disease) [2]. Traditionally, *Ayurveda* recognizes three main energies (*doshas*) of the body - *pitta* (metabolism), *kapha* (structure) and *vata* (mental). Set at birth at various ratios to maintain physiological homeostasis, their imbalance is thought to cause mental illnesses [5,6]. Among the three *doshas*, one is typically dominant in each patient, and such individual constitution (*prakriti*) determines the susceptibility to specific disorders, based on physical, psychological, physiological and behavioral traits that are independent of social, ethnic and geographical variables [7].

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For millennia, *Ayurveda* has successfully treated various human central nervous system (CNS) disorders, including anxiety, depression, cognitive decline, as well as Alzheimer's (AD), Huntington's (HD) and Parkinson's diseases (PD, [Tables 1](#) and [1-Supplementary material](#)). Complementing clinical studies, experimental animal models, especially laboratory rodents, are increasingly utilized in translational *Ayurveda* research [6], including studying CNS effects of *Ayurvedic* medicine (see further). Mounting evidence also suggests that alternative model species, such as zebrafish (*Danio rerio* Hamilton), can also be a useful tool to study effects of *Ayurvedic* medicines on CNS functions [8,9]. To examine this further, here we discuss how CNS effects of various *Ayurvedic* medicinal plants can be elucidated using zebrafish models, based on a comprehensive search of published literature from all major biomedical databases (Web of Science, PubMed, ScienceDirect, Scopus, and Google Scholar). We also evaluate the existing challenges and limitations of using fish models for studying *Ayurvedic* pharmacotherapy, as well as outline future directions of translational research in this field.

2. Selected examples of experimental models to study CNS effects of *Ayurvedic* medicine

As already noted, in addition to rich clinical data, multiple animal (experimental) models have been used to study *Ayurvedic* medicines [6]. For example, Indian ginseng (*Withania somnifera* Linnaeus) that is efficient in managing stress, epilepsy and PD clinically [10], exhibits positive effect in experimental AD models [2], reducing neuro-fibrillary tangles and scavenging free radicals to protect dopaminergic neurons in rodents [11]. Known as *ashwagandha* and used in *Ayurveda* for millennia to treat pain, inflammation, insomnia and stress, this plant provides an extract that reduces lipid peroxidation, upregulates tyrosine hydroxylase (TH) in substantia nigra and exerts anti-oxidant properties in mice [10,12]. In transgenic AD mouse model, its root extracts show neuroprotective effects against amyloid beta (A β) and peroxide toxicity, and decrease A β formation in the brain [13]. Promoting neurite regeneration and increasing central gamma-aminobutyric acid (GABA) tone, this plant is beneficial in epilepsy [14], also improving cognition, restoring motor function and controlling acetylcholine esterase (AChE) activity in an experimental HD model [15]. Methanolic and aqueous extracts of Indian ginseng also promote serotonergic transmission via the 5HT_{2A/2C} receptor in the mouse model of the obsessive compulsive disorder [16].

Another popular plant, water hyssop (*Bacopa monnieri* Linnaeus Wettstein), known in *Ayurveda* as *brahmi* ('plant of Lord Brahma'), is widely used to treat cognitive deficits, anxiety and epilepsy clinically [17]. This plant contains several saponins (bacoside A3, bacoside II, bacosaponin C and bacoside I) that determine its therapeutic effects and biological activity in vivo. For example, recommended for human AD and PD patients [18], water hyssop also improves cognition and reduces anxiety and stress in animal models [5], and promotes neuroplasticity by increasing the expression of brain-derived neurotrophic factor (BDNF) in mice [19]. Lower dose of this plant extract reduces

inflammatory interleukins in microglial cells [20], also decreasing oxidative stress, protecting neuronal and glial cells, and lowering neuro-inflammation in rats [21,22], in addition to its well-reported cognitive enhances activity in humans [17].

Curcumin is an active compound of turmeric (*Curcuma longa* Linnaeus), promoting neuroprotection in AD and exerting anti-inflammatory, anti-oxidant (and anxiolytic) activity in mouse models and in vitro [3]. In *Ayurveda*, this plant is used as an anti-amyloidogenic and anti-inflammatory agent, also showing neuroprotective action in tardive dyskinesia, AD, depression, epilepsy and other CNS disorders [23]. Inhibiting monoamine oxidase (MAO)-A in mice, curcumin elevates brain monoamines [24] in synergistic interactions with serotonergic antidepressants [25]. Currently approved by the US Food and Drug Administration (FDA) in Phase II clinical trials against AD as Longvida [26], it also reduces the activation of microglia and astrocytes, as well as cytokine production and the nuclear factor kappa-light-chain enhancer of activated B cells (NF κ B) signaling pathway, all suggesting beneficial activity against AD, in addition to its clinical use as a complementary treatment for neuropathic pain [27].

Indian pennywort (*Centella asiatica* Linnaeus) is another commonly used *Ayurvedic* plant [28] with well-known potent antimicrobial, anti-oxidant, antiulcer, anti-diabetic, anticancer, antidepressant and nootropic clinical activity [29]. Mentioned in *Ayurveda* as *gotu kola* or *mandukaparni*, and rich in flavonoids, antioxidants and vitamins, Indian pennywort has been traditionally used to improve mindfulness and cognitions, as well as an antiepileptic, antidepressant, wound healing and antiseptic therapy. Representing a major *Ayurvedic* 'Medhya Rasayana' (literally - 'improving mind', nootropic) plant, it also improves memory and learning in rodents [30,31], inhibits phospholipase A2 (PL-A2) and acetylcholine esterase (AChE), protects cerebellum from A β -induced neurotoxicity [32], evokes anti-inflammatory activity in lipopolysaccharide (LPS)-treated rat neurons, and possesses anxiolytic-, anti-stress and antidepressant-like effects in vivo [33].

Morning glory (*Convolvulus pluricaulis* Linnaeus, another major *Medhya Rasayana*/nootropic *Ayurvedic* plant known as *shankhapushpi*) and spikenard (*Nardostachys jatamansi* Linnaeus, a promnestic, antidepressant and antiepileptic plant) both exert nootropic effects clinically [2] and in rodent models [34]. Likewise, Asian pigeonwing (*Clitoria ternatea* Linnaeus) root extract improves rat passive avoidance learning and retention. A traditional *Ayurvedic* medicine *aparajita*, it has long been used as a nootropic, anxiolytic, antidepressant, anticonvulsant and sedative agent. Commonly known as 'butterfly pea', this plant generates multiple metabolites (e.g., triterpenoids, glycosides, anthocyanins and steroids) that contribute to its biological activity in vivo [35].

Considered one of the most important *Ayurvedic* herbs, tulsi (*Ocimum sanctum* Linnaeus) promote general rejuvenation of mind and body, improves cognition and metabolism [36], reduces stress, anxiety and depression [37], and exerts anticonvulsant activity clinically [38]. Similarly to humans, ethanol extract of tulsi evokes an anti-stress profile in rodents, reducing plasma corticosterone responses to acute or chronic

Table 1
Selected *Ayurvedic* herbs, their active compounds and various CNS effects.

<i>Ayurvedic</i> plants	Botanical name	Major active compounds	Targeted brain conditions	References
Curcumin	<i>Curcuma longa</i>	Curcumin	Alzheimer's (AD) and Parkinson's (PD) diseases, anxiety	[2,3]
Brahmi	<i>Bacopa monnieri</i>	Bacoside A, bacoside B and bacosaponins, D-mannitol	AD, PD, anxiety and stress	[21,22]
Ashwagandha	<i>Withania somnifera</i>	Withanolides A, B, Q and IV, V, somnifene, somniferine and sominone	AD, PD and Huntington's disease (HD), stress and epilepsy	[11,14]
Shatavari	<i>Asparagus racemosus</i>	Shatavarin IV	AD, PD and neuroinflammation	[2]
Shankhapushpi	<i>Convolvulus prostratus</i>	Glycosides, flavonol, anthocyanins and steroids	Anxiety, mental fatigue, and insomnia	[97,98]
Jatamansi	<i>Nardostachys jatamansi</i>	Sesquiterpenoids and valeriananoids	AD and insomnia	[99,100]
Mandukaparni	<i>Centella asiatica</i>	Asiaticoside A, B and asiatic acid	AD and PD	[101]
Gugul	<i>Commiphora wightii</i>	Guggulsterol (I-III, IV, V) and E,Z-guggulsteron	Dementia	[102]

Please see text and [Table 1-Supplementary material](#) for a complete list of *Ayurvedic* plants mentioned here.

noise stress, and improving memory [39]. The lipid peroxidase activity, deficits in active avoidance learning and retention of learned behavior are reversed by tulsi extract in the rat AD model, also normalizing superoxide dismutase activity [40].

The role of brain–gut axis has been increasingly recognized in pathogenesis of various CNS disorders, including depression, anxiety and neurodegenerative disorders [41]. For example, the irritable bowel disease (IBD), causing a chronic inflammation of the gastrointestinal tract and causing both Crohn disease and ulcerative colitis, also increases CNS inflammatory responses, further contributing to affective disorders clinically [42]. Crohn patients have higher anxiety scores than healthy individuals [43], and are more prone to depression [44]. Various *Ayurvedic* medicines have long been used to correct the brain-gut axis clinically [45]. For instance, the curcumin extract (Curcugen) administered for 8 weeks at a dose of 500 mg once daily is associated with greater improvements in digestive complaints and anxiety levels in adults with self-reported digestive complaints [46]. Subjects treated with another *Ayurvedic* medicine, ashwagandha (*Withania somnifera* L. Dunal), for 8 weeks demonstrate significant improvements compared with the placebo group in cognitive evaluation (e.g., immediate and general memory) [47]. In addition to clinical finding, rodent models show reduced anxiety- and depression-like behavioral deficits following their treatment with *Ayurvedic* medicine (e.g., *Withania somnifera* [48], aqueous extract of *Azadirachta indica* A. Juss [49], probiotics associated with curcumin [50]). Collectively, these findings strongly support the growing value of animal models for studying various aspects of CNS effects of *Ayurvedic* medicine.

3. Assessing selected *Ayurvedic* medicines in zebrafish CNS models

Complementing clinical and rodent findings, the zebrafish has recently emerged as a powerful vertebrate *in-vivo* system for CNS disease modeling and drug screening [51]. For example, both adult and larval zebrafish represent powerful vertebrate *in-vivo* model systems to study CNS functions [52] due to their high genetic and physiological similarity to humans, rapid development and shared neurotransmitter systems [53]. Other advantages of zebrafish models include neuromorphological similarity to humans [54], easy maintenance, high fecundity [55] and the availability of well-established behavioral assays. Zebrafish are also sensitive to all major classes of CNS drugs, offer cost-effective model for *in vivo* drug screening and show potential for high-throughput screening (HTS) [56].

Recognizing its potential for bioscreening various medicines, the zebrafish has recently become utilized as a preclinical model for obtaining insights into molecular mechanisms of action and for

developing novel treatments for neurological disorders based on *Ayurvedic* herbal medicine [57]. Mounting evidence, summarized in Table 2 and including selected examples discussed further in detail, strongly supports the utility of zebrafish in studying CNS effects of *Ayurvedic* plants. Fig. 1 depicts examples of CNS effects.

3.1. Indian pennywort

Indian pennywort has been extensively tested in zebrafish. For example, in rotenone neurotoxicity assay, when co-incubated with the extract of Indian pennywort, adult zebrafish elevate BDNF levels in midbrain, their dopaminergic neurons become protected from toxicity, and fish display reduced PD-like phenotypes (e.g., lower alpha-synuclein aggregation, increased locomotion and higher brain dopamine levels) [28]. The active constituents of this herb (e.g., brahmoside and brahminoside) are traditionally used for CNS relaxation in humans and exert similar effects in animals [58], whereas asiaticoside, its another active compound, reduces hydrogen peroxide-induced cell death, free radical concentrations *in-vitro* and A β levels in rat hippocampus [59].

3.2. Indian ginseng

Indian ginseng extract exposure attenuates benzo[a]pyrene neurotoxicity in adult zebrafish, increasing brain antioxidant status and restoring normal scototaxis (dark preference) behavior [60]. Interestingly, in *Ayurveda* practice, this plant is commonly used as an anxiolytic, nootropic [11], anti-PD/AD/HD and anti-epileptic agent [10], and its effects in zebrafish strikingly parallel clinical findings.

3.3. Curcumin and carob

Curcumin improves locomotor performance in adult zebrafish rotenone-based PD models [61], whereas leaf extract of tannins-rich carob (St. John's bread, *Ceratonia siliqua* Linnaeus) improves cognitive functions and lowers anxiety in adult zebrafish PD model evoked by a neurotoxin 6-hydroxydopamine (6-OHDA) – the effects likely mediated by restoring brain antioxidant status and regulating AChE activity [62].

3.4. Tulsi and areca palm

Paralleling clinical nootropic use of tulsi, its leaf extract reduces memory impairment in adult zebrafish caused by scopolamine, an anticholinergic drug commonly used to induce memory deficits [63]. Known as *puga* in *Ayurveda*, extracts from leaves and nuts of areca palm (*Areca catechu* Linnaeus) are commonly used clinically to treat ulcers, pain and indigestions. Arecoline, the main psychoactive compound of

Table 2
Selected examples of using zebrafish models for screening CNS effects of *Ayurvedic* medicines.

Medicines	Botanical name	Sex	CNS effects	References
Indian pennywort	<i>Centella asiatica</i>	Both	Protected dopaminergic neurons from rotenone toxicity by increasing BDNF levels and reduced α -synuclein aggregation	[28,103]
Indian ginseng	<i>Withania somnifera</i>	Both	Improved brain antioxidant status and neuroprotection against benzo[a] pyrene toxicity	[60]
Suvarna Bhasma ^a	-	Not specified	Reduced behavioral deficits induced by rotenone administration (a PD model)	[70]
Curcumin	<i>Curcuma longa</i>	Both	Reduced anxiety-like behavior	[71]
Tulsi	<i>Ocimum tenuiflorum</i>	Both	Improved locomotion and reduced rotenone toxicity in PD model	[61]
		Male	Protected memory against scopolamine	[63]
Asian pigeonwing	<i>Clitoria ternatea</i>	Both	Reduced stress responses induced by reserpine	[65]
Brahmi	<i>Bacopa monnieri</i>	Not specified	Reduced anxiety-like behavior and cortisol levels following exposure to stress stimuli	[66]
Carob	<i>Ceratonia siliqua</i>	Both	Antioxidant and anti-AChE activity, improved cognitive function in the 6-OHDA-induced PD model	[62]
Neem	<i>Azadirachta indica</i>	Not specified	Reduced locomotion and increased anxiety-like behavior	[104]

^a gold nanoparticles.

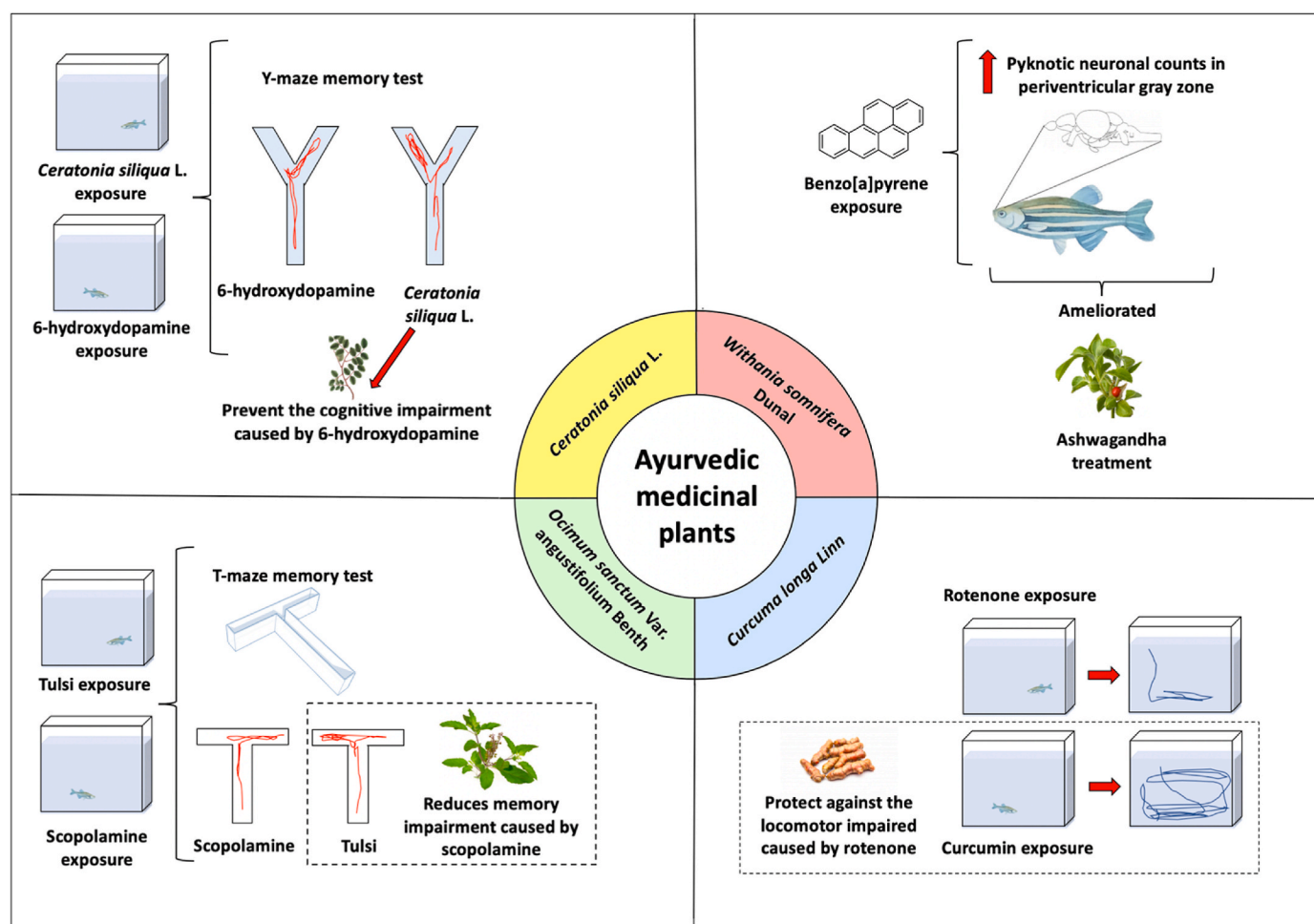


Fig. 1. Selected examples of CNS effects of Ayurvedic medicinal plants in zebrafish models. *Ceratonia siliqua* L. (carob) is a Mediterranean medicinal plant prevents the cognitive impairment of 6-hydroxydopamine (6-OHDA, a Parkinson's disease (PD) model) [62]. Ashwagandha (*Withania somnifera* Dunal) leaf extract ameliorated the increase in the pyknotic neuronal counts in periventricular gray zone caused by benzo[a]pyrene neurotoxicity [60]. The extract of tulsi leaves reduces memory deficits caused by scopolamine in zebrafish [63]. Curcumin, the main ingredient of turmeric, protects against the locomotor impaired caused by rotenone (a PD-like model) [61].

areca (betel) nuts and their water extract, have been recently studied in zebrafish, whose CNS responses parallel clinical and rodent findings, and show pronounced anxiolytic-like behavior and elevated norepinephrine and serotonin levels [64]. On the one hand, by showing similar effects across taxa, these data further support evolutionary conservation of psychoactive and neurochemical effects of Ayurvedic medicines across vertebrates. On the other hand, although anxiolytic effects of puga are not listed as its major use in Ayurveda, the fact that consistent zebrafish, rodent and clinical CNS phenotypes are evoked by arecoline [64] suggests that repurposing Ayurvedic medicines, for example, based on their zebrafish-based CNS screening, may also lead to novel potential pharmacotherapies for various brain disorders.

3.5. Asian pigeonwing

Asian pigeonwing, a potent stress-reliever and brain stimulant in humans and rat models, has also been tested in zebrafish [65]. In the fish model of pharmacogenic affective syndrome, the clitorienolactone- and isoflavonoid-rich extract from the roots of this plant, reduces motor deficits (freezing and hypolocomotion) induced by a dopamine-depleting drug reserpine [65], hence strikingly paralleling brain stimulant action of this popular Ayurvedic medicine in humans.

3.6. Water hyssop

Water hyssop, traditionally used in Ayurveda as a neural tonic, potently reduces thigmotaxic and scototaxic behavior (anxiolytic effect) in zebrafish larvae exposed to osmotic stress as compared to untreated stressed larvae, as well as reduced the stress-induced release of cortisol in zebrafish larvae, suggesting that this plant may be an option for managing stress and anxiety [66].

3.7. Satawari and black myrobalan

Given the success of Ayurvedic medicines in targeting the brain-gut axis [45], and reflecting the growing potential of zebrafish in modeling this axis [67], some compounds have been tested in zebrafish models, aiming to better understand the gut-brain axis, as well as its influence on CNS disorders. For example, in a zebrafish model of IBD, methanol extracts of Indian plants water roof (*Asparagus racemosus* Willdstein) and especially black myrobalan (*Terminalia chebula* Retzius), known as *shatavari* and *haritaki* in Ayurveda, lower inflammatory cells in blood and improve pathological scores in the gut compared with positive prednisolone control [68]. In sum, these findings support zebrafish as useful model species to study the effects of Ayurvedic medicinal plants on inflammatory and CNS disorders, as well as the development of novel therapies for targeting gut-brain axis [67].

3.8. Suvarna Bhasma

In addition to various herbal medicines, other types of *Ayurvedic* treatments can also be tested in zebrafish. For instance, believed to have rejuvenating, nootropic and stress-reducing properties in *Ayurveda*, nano-sized gold particles, *Suvarna Bhasma* (SB), have a powerful potential to treat neurological diseases (e.g., PD) clinically [69]. In a zebrafish PD model (induced by rotenone administration), behavioral deficits are reduced by SB treatment, which also prevents dopamine deficits in fish brain, paralleling neuroprotective effects of SB against PD [70]. Furthermore, utilizing non-toxic doses of SB in zebrafish produced beneficial (anxiolytic-like) behavioral effects in the novel tank test, also strikingly resembling well-established calming, anti-stress effects of this medicine in humans [71].

4. Prospects of zebrafish models in neuroscience and *Ayurvedic* drug screening research

Analyses of reported CNS effects of *Ayurvedic* plants in animal models (Table 1-Supplementary material) reveal predominantly promnestic (nootropic) and anti-stress effects in vivo, which are generally in line with clinical disorders treated with *Ayurvedic* herbs (Table 1). However, identifying novel clinical effects of traditional *Ayurvedic* plants may also be necessary, especially targeting other illnesses, such as neurodevelopmental disorders like autism spectrum disorder, currently lacking the specific drug therapy [72]. *Ayurvedic* drugs have already shown its efficacy in the treatment of pain disorders as strong analgesic and anti-inflammatory agents [73], and may also be used to treat complex multifaceted brain disorders, such as schizophrenia [74], where, for example, water hyssop extracts given chronically improve positive and negative clinical symptoms of psychoses without overt adverse effects [75]. As zebrafish become a valuable model organism to mimic schizophrenia [76] and other related CNS conditions, testing this possibility in fish becomes possible.

Importantly, the effectiveness of *Ayurvedic* medicinal plants depends on the bioavailability of their active compounds over a sustained period of time [77]. Addressing this aspect, recent nanotechnologies have been developed to enhance bioavailability and bioactivity of *Ayurvedic* medicines [77]. For instance, micro and nano-carrier-mediated drug delivery systems [78] can prolong biological effects of the drugs in the action site by reducing its clearance and increasing therapeutic effects [79]. Thus, capitalizing on home-tank water immersion-based drug administration, similar long-term treatment approaches can be developed in zebrafish, to further enhance their CNS models and screens for *Ayurvedic* medicines. The supercritical fluid micronization technology represents another useful tool to increase drug bioavailability [80] and therapeutic efficacy [81]. For example, micronized resveratrol shows promising effects in an epilepsy model, preventing tonic/clonic seizures in adult zebrafish [82]. Similar approaches can also be used to enhance therapeutic effects of *Ayurvedic* medicines in this and other CNS disease models using zebrafish.

Various intrinsic factors (e.g., sex, individual, anatomic, behavioral and physiological differences) modulate pharmacological responses in clinical and preclinical studies [83]. For instance, as body temperature modulates drug effects in humans and rodents [84], this factor can differentially impact *Ayurvedic* therapies in poikilothermic zebrafish (typically maintained in laboratories at 25–28 °C) and homeothermic mammals (typically maintaining a 35–37 °C body temperature). Some extrinsic factors can also differentially influence behavioral and CNS drug response in preclinical studies across species, as, for example, experimenter's sex impacts such responses in rodents [85], but not in zebrafish [86].

For large-scale *Ayurveda* studies, zebrafish high-throughput imaging assay can be developed for drug neurotoxicity screening, to better understand drug pharmacokinetics and pharmacodynamics. For instance, *Ayurveda* utilizes a tedious procedure for the preparation of *rasasindura*

(a mercury-based medicinal formulation), which in zebrafish larvae, even at high concentrations of 1000 ppm, did not show toxicity or morphological changes, suggesting that this *Ayurvedic* medicine can be non-toxic in vivo up to a reasonable concentration [87]. While many experiments with *Ayurvedic* drugs involved adult fish (Table 2), zebrafish larvae is another promising tool since their small size enables placing them into multi-well plates filled with only 200 µl of fluid, requiring few milligrams of compounds for screening [88]. Although identification of bioactive compounds from *Ayurvedic* plants normally requires multiple chromatographic procedures and large amounts of material, which is problematic in larval assays, novel methods for larval behavioral testing with micro-fractionation techniques for bioactive compound identification from traditional herbal medicine have been developed [89].

An interesting line of research also involves the *rasa shastra* branch of *Ayurveda* that combines traditional Indian medicinal plants with inorganic components, mostly metals. For example, the combination of *Ayurvedic* plant extracts with metals may have some therapeutic potential in zebrafish models [8], meriting further scrutiny. Finally, the interaction of *Ayurveda* medicines with conventional drugs is also of clinical and translational interest. For example, Indian ginseng potentiates antiepileptic activity of diazepam [90], gugul has synergistic effect with hypolipidemics in humans, whereas tulsi is hepatotoxic when co-administered with paracetamol in rats [91]. Given the value of zebrafish in drug screening, especially HTS [92], the use of such sensitive models to probe CNS interactions between *Ayurvedic* medicines, and/or their interaction with other (e.g., allopathic) drugs, becomes critically important.

5. Future directions

While there is a growing support for assaying CNS effects of various *Ayurvedic* medicines in zebrafish, their underlying molecular mechanisms remain largely understudied, complicating phenotypical translation and their comparison with other drugs, in terms of both efficiency and targets. It is especially important given multiple *Ayurvedic* plants with potential CNS effects, and the fact that many representatives of different plant families exert similar (e.g., promnestic and anxiolytic) profiles in clinically or animal models (Table 1, 1-Supplementary material, and 2). Together, this necessitates further studies into potential shared neurobehavioral domains targeted by *Ayurvedic* medicines across taxa. For example, nootropic and anxiolytic CNS effects in animal models are prevalent among *Ayurvedic* medicines (Fig. 2), correlating with their commonly reported antioxidant, anti-inflammatory and neuroprotective activity in vivo (Tables 1 and 1-Supplementary material). Taken together, this raises the possibility that behavioral and molecular effects of *Ayurvedic* medicines may be interrelated. Hence, this not only suggests some shared molecular pathways underlying beneficial CNS effects of *Ayurvedic* medicine, but also implies that cognitive/behavioral responses may be due to improved neuroprotection and anti-oxidant effects.

Moreover, only relatively few *Ayurvedic* plants have been tested for CNS research in zebrafish models presently (Table 2), resulting in a lack of critical mass of empirical evidence in the field. The use of *Ayurvedic* plants extract in different sexes, strains and ages of zebrafish is also a challenge since different fish may respond differently to anxiolytic compounds and drugs that affect their general locomotor activity. With the growing number of *Ayurvedic* plants with well-established CNS effects (Table 1-Supplementary material), further research is needed to fill this knowledge gap utilizing zebrafish models.

6. Potential challenges and limitations

In general, treating brain disorders is complicated by low drug bioavailability and ineffective delivery, unclear genetic risk factors, often unsuitable model organisms for drug discovery and failure to use

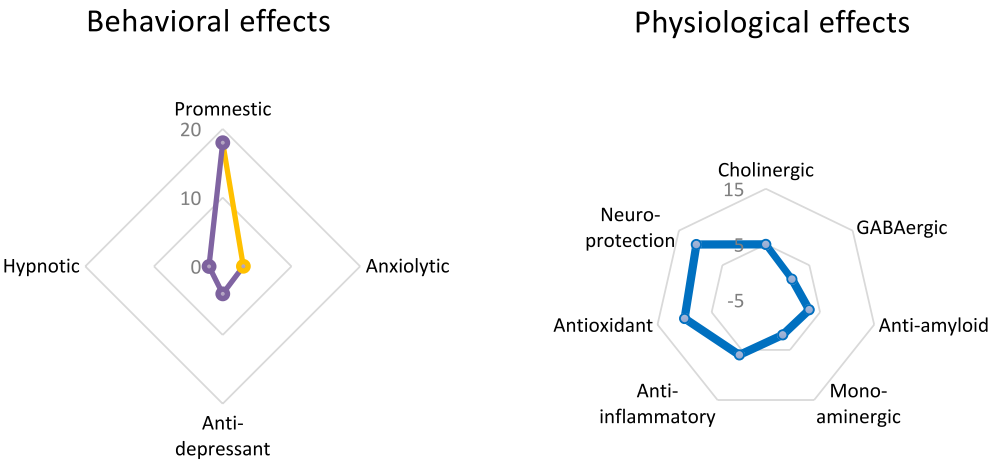


Fig. 2. Summary of common behavioral and physiological CNS effects involved in preclinical activity of *Ayurvedic* medicines. Data represent the occurrence of specific effects in plans discussed here and listed in [Table 1-Supplementary material](#). GABA - gamma aminobutyric acid.

individual (‘personalized’) medicine-based approaches. As various animal models become used for screening *Ayurvedic* drugs ([Tables 1 and 2](#)), there are also some challenges and limitations in this field, including problematic unavailability of standards for raw materials, identification of bioactive compounds and pharmacognostic analyses, low bioequivalence and safety concerns. The dose standardization for bioactive molecules from *Ayurvedic* plants is also an important, yet often neglected consideration, especially since animal doses often differ from traditional doses recommended clinically.

Potential adverse effects of *Ayurvedic* herbal plants also present a concern, especially since few studies have explored them in humans and animals. For example, water hyssop causes poor fertility in rats [93], but not in humans, whereas Indian pennywort may cause severe neurological problems at high doses, including headache, nausea, dizziness and extreme drowsiness [58]. Thus, clinical trials are needed to confirm biosafety profile of *Ayurvedic* plants, as well as their comparative studies with allopathic medicines, to evaluate their efficacy and safety profiles for the use in humans. Furthermore, herbal constituents undergo in vivo biotransformation, which may lower the efficacy of *Ayurvedic* plants, and can differ between fish, rodents and humans. There are also difficulties in assessing active molecules in pharmacokinetic studies [94],

especially since the crude raw herbal material may differ for geographical locations, climatic conditions, harvesting methods and/or collection protocols, hereby complicating standardizing the drug treatment.

7. Conclusion

Ayurvedic medicines represent affordable and effective folk medicines that may lead to identifying novel drug targets and the discovery of structurally and functionally novel drugs for human brain disorders. The use of zebrafish models is growing rapidly in CNS drug discovery [51], including *Ayurvedic* medicine ([Table 2](#)), albeit not without translational and practical challenges discussed above. Multiple open questions also remain in this field ([Table 3](#)), warranting further studies. For example, to what extent zebrafish physiology recapitulates clinical therapy? Are there reliable, evolutionarily conserved biomarkers relevant to therapeutic potential of *Ayurvedic* medicines that are shared between fish and humans? Does *Ayurvedic* therapy exert longer-term genomic, epigenetic or epigenomic effects in zebrafish brain? And, finally, can modern nanotechnologies enhance the efficacy of *Ayurvedic* medicines in zebrafish CNS models? For instance, nanomedicine is increasingly used

Table 3
Selected open questions related the use of *Ayurvedic* plants in zebrafish model.

Questions
• What are mechanisms involved in zebrafish CNS responses to different <i>Ayurvedic</i> drugs? • Are there sex-, strain- and age differences in CNS effects of <i>Ayurvedic</i> plants in zebrafish? • What are pharmacological and toxicological doses of <i>Ayurvedic</i> plant extracts in zebrafish models? • Do different routes of administration of <i>Ayurveda</i> drugs affect screening data in zebrafish assays? • Does different metabolic system of zebrafish (vs. Humans) affects the efficacy of <i>Ayurvedic</i> drugs? • How similar are behavioral effects of <i>Ayurvedic</i> plant extracts in zebrafish vs. odents and humans? • What are genomic and epigenomic CNS responses to <i>Ayurvedic</i> drugs in zebrafish models? • Are there transgenerational (e.g., epigenetic) effects of <i>Ayurvedic</i> drugs in zebrafish? • What are the effects of short-, medium- and long-term of <i>Ayurvedic</i> plant extract exposure on zebrafish behavioral (e.g., aggressiveness, anxiety, motor activity) and cognitive phenotypes? • What are effects of <i>Ayurvedic</i> plant extract on zebrafish gut microbiota, and how likely can they impact fish behaviors? • Can zebrafish CNS models be used to study drug-drug interactions for <i>Ayurvedic</i>, as well as <i>Ayurvedic</i> and other medicines (e.g., antidepressants, anesthetics)? • How to tackle the problems with blood brain barrier (BBB) permeability of <i>Ayurvedic</i> compounds? • How do different extraction methods of <i>Ayurvedic</i> plants affect their efficacy in zebrafish CNS models? • How do zebrafish and mammalian neuroglia interact with <i>Ayurvedic</i> drugs active compounds? • How can we use <i>Ayurvedic</i> remedies in neurodevelopmental studies in zebrafish? Are there any delayed CNS effects of neonatal exposure to <i>Ayurvedic</i> medicines? • How can temperature factors (e.g., body or environmental temperature) affect the efficacy of <i>Ayurvedic</i> medicinal therapies in zebrafish vs. clinical studies? • What are toxic side-effects of <i>Ayurvedic</i> medicines in zebrafish? What are other side-effects (both mental and non-mental) produced by <i>Ayurvedic</i> medicines? • How do <i>Ayurvedic</i> medicines affect glial cells in zebrafish? • What are proteomic and metabolomic alterations associated with acute and chronic exposure to <i>Ayurvedic</i> medicines in zebrafish? • Are there placebo and/or nocebo CNS effects of <i>Ayurvedic</i> medicines? If yes, can these drug responses be modeled in zebrafish, and what are their potential physiological mechanisms? • How can modern nanotechnologies enhance the efficacy of <i>Ayurvedic</i> medicines in zebrafish CNS models?

in drug development, reducing damage to healthy cells and optimizing the consumption of drug dosage [95]. Thus, incorporating nanotechnology in traditional *Ayurveda* may eliminate one of the major pitfalls of metal toxicity by converting treatments into biological nanoparticles, as in SB and other *bhasmas* - *Ayurvedic* preparations using incinerated metals [96]. Testing this and other *Ayurvedic* medicines in zebrafish will support the growing utility of these fish in probing *Ayurvedic* drug therapy and studying potential molecular mechanisms of its CNS effects.

CRedit authorship contribution statement

Pallab Chakraborty: Conceptualization, Writing – original draft, Writing – review & editing. **Subharthi Pal:** Conceptualization, Writing – original draft, Writing – review & editing. **David S. Galstyan:** Writing – original draft, Writing – review & editing. **Elena V. Petersen:** Writing – original draft, Writing – review & editing. **Tamara G. Amstislavskaya:** Writing – original draft, Writing – review & editing. **Tatiana O. Kolesnikova:** Writing – original draft, Writing – review & editing. **Murilo S. de Abreu:** Conceptualization, Writing – original draft, Writing – review & editing. **Allan V. Kalueff:** Conceptualization, Writing – original draft, Writing – review & editing, Funding acquisition, Supervision.

Declaration of generative AI in scientific writing

None of such technologies were utilized during the preparation of this manuscript.

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Conflict of interest

Authors declare no conflicts of interest.

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Appendix A. Supplementary data

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